

Probabilistic uncertainty for CFD surrogates

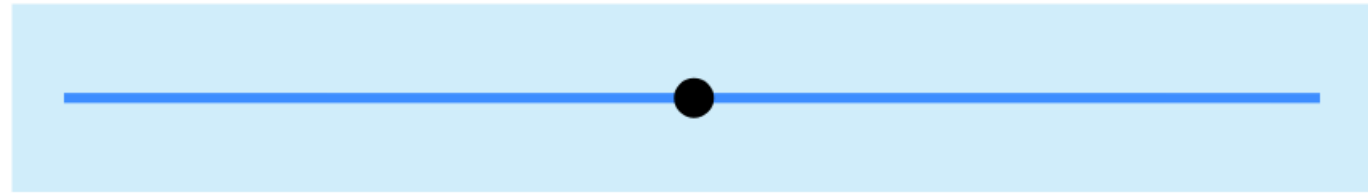
From a loss function to a calibrated predictive distribution

A point estimate cannot express confidence

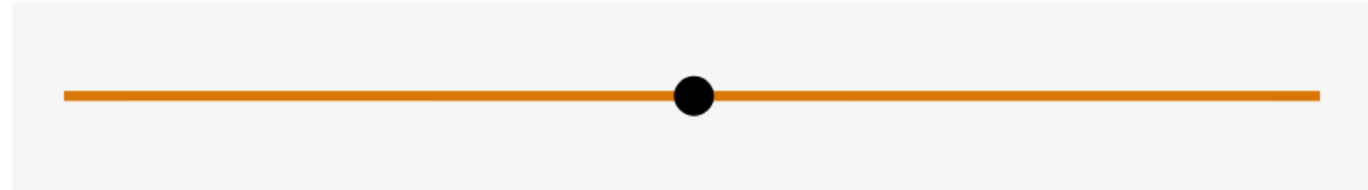
SAME MEAN ERROR, DIFFERENT SCIENTIFIC RISK

$$\hat{y}(x)$$

One best guess



narrow distribution



wide distribution

The mean can be identical while the decision should change.

Week 2.1 extends supervised regression

Week 2 already gives us

features and targets
scaling
losses
case-wise splits

This increment adds

- 1 write an observation model
- 2 compute posterior prediction
- 3 score the full distribution
- 4 calibrate without touching blind cases

No new neural architecture is required.

Uncertainty has three different causes



ALEATORIC

Noise or unresolved variability

More data may not remove it.



EPISTEMIC

Limited knowledge of the map

More informative data can reduce it.



DISCREPANCY

The simulator misses reality

A confident surrogate can still be wrong.

Do not collapse these mechanisms into one unexplained error bar.

A likelihood turns loss into a modeling choice

$$y = f\theta(x) + \varepsilon$$

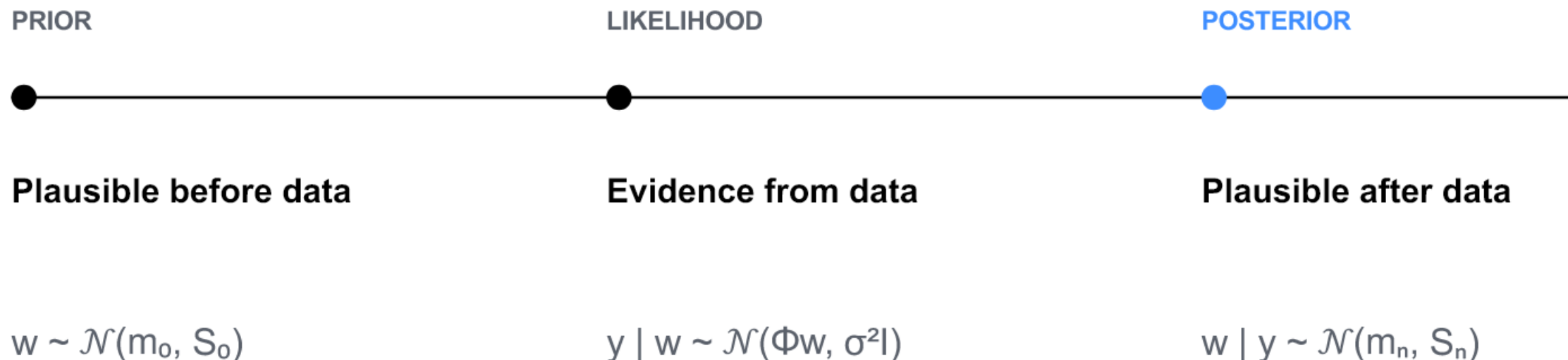
$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

GAUSSIAN OBSERVATION MODEL

$$\begin{aligned} & -\log p(y \mid \theta, \sigma^2) \\ & = \text{constant} + \sum (y - f\theta(x))^2 / 2\sigma^2 \end{aligned}$$

MSE is therefore not neutral: it encodes constant-variance Gaussian residuals.

Bayesian linear regression is an exact update



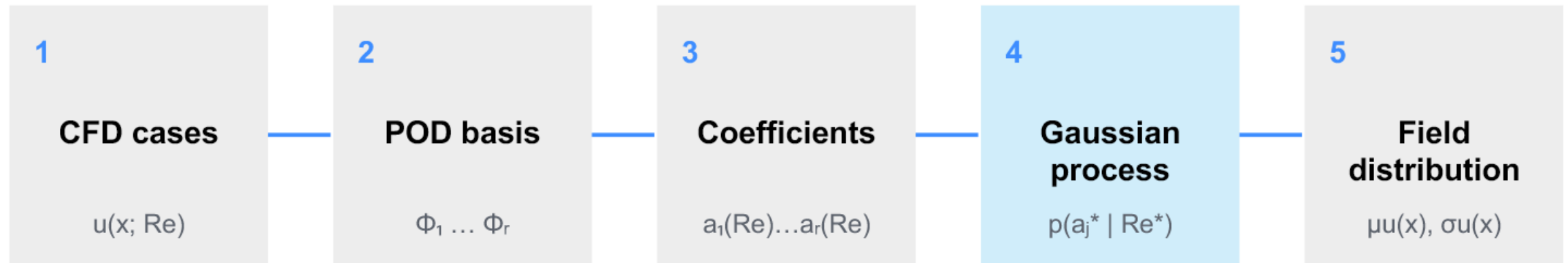
Posterior precision = prior precision + data precision

Prediction uncertainty has two terms

$$\text{Var}[y^* \mid \mathbf{x}^*, D] = \underbrace{\sigma^2}_{\text{observation noise}} + \underbrace{\boldsymbol{\phi}^{*T} \mathbf{S}_n \boldsymbol{\phi}^*}_{\text{parameter uncertainty}}$$

Report which terms are included. A posterior over weights does not automatically include simulator discrepancy.

POD compresses field uncertainty



Low-rank structure reduces a spatial field to a few probabilistic coefficient functions.

Calibration must stop before the blind gate

DEVELOP



Fit POD + GP

Re = 100, 150, 200, 225, 250,
350, 400

CALIBRATE



**Scale interval width
once**

Re = 300

BLIND TEST



Measure without tuning

Re = 175, 275, 375

The split unit is a complete Reynolds-number case—not individual grid nodes.

Judge accuracy, calibration, and sharpness

Optimizing only one can produce a useless distribution.

01

Accuracy

Is the predictive mass near truth?

02

Calibration

Do nominal levels match frequency?

03

Sharpness

Are intervals as narrow as justified?

Score the distribution, not only its mean

GAUSSIAN NEGATIVE LOG LIKELIHOOD

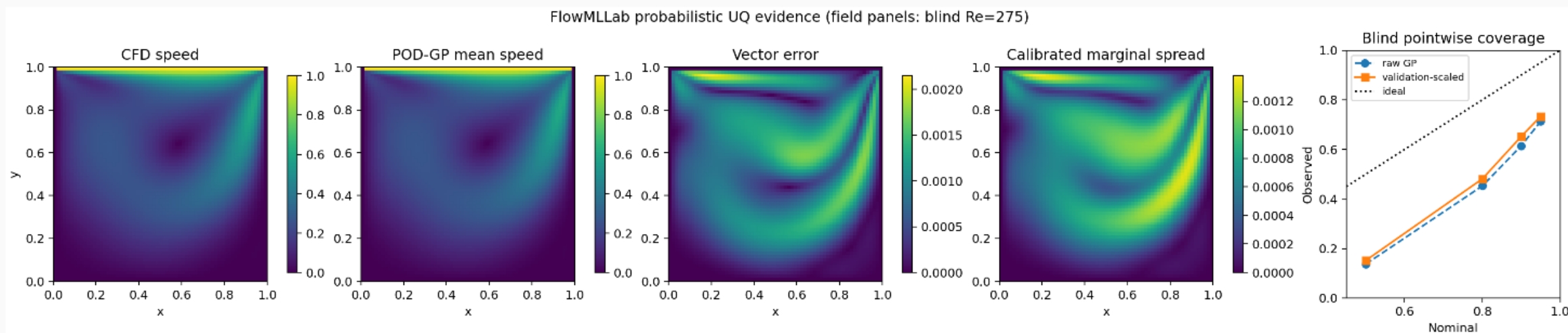
$$\text{NLL} = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2}(y-\mu)^2/\sigma^2$$

CONTINUOUS RANKED PROBABILITY SCORE

CRPS compares the predictive CDF with the observed step CDF.

Lower is better. Both scores penalize confident errors.

Blind evidence beats a polished uncertainty story



The mean improved; coverage did not

Validation-only scaling increased blind 90% coverage, but did not approach the nominal target.

0.3333%

Mean POD–GP relative L_2 error

0.414%

Mean interpolation relative L_2 error

65.2%

Calibrated coverage at nominal 90%

Make the uncertainty claim as auditable as the mean

- ❑ State what is random
- ❑ Keep physical cases intact
- ❑ Freeze calibration before blind testing
- ❑ Report accuracy, coverage, and width
- ❑ Preserve under-coverage as evidence

Exit prompt

What new data would separate epistemic uncertainty from CFD model discrepancy?